

2D Ising Model Simulation

Metropolis Monte Carlo

The 2D Ising model is a simple mathematical model used to study how magnetic materials behave at different temperatures. In this model, a grid of atoms (called spins) can either point up or down, and neighbouring spins interact with each other.

We simulated this model on a 50x50 grid using the Metropolis Monte Carlo method, which randomly flips spins and accepts or rejects the flip based on energy rules. The simulation was run at five different temperatures — $T = 1.0, 1.8, 2.269, 3.0,$ and 4.0 — to observe how the system behaves when it is cold (ordered), hot (disordered), and exactly at the critical temperature $T_c = 2.269$, where a phase transition occurs.

At each temperature, 50 independent runs were performed and the following quantities were measured:

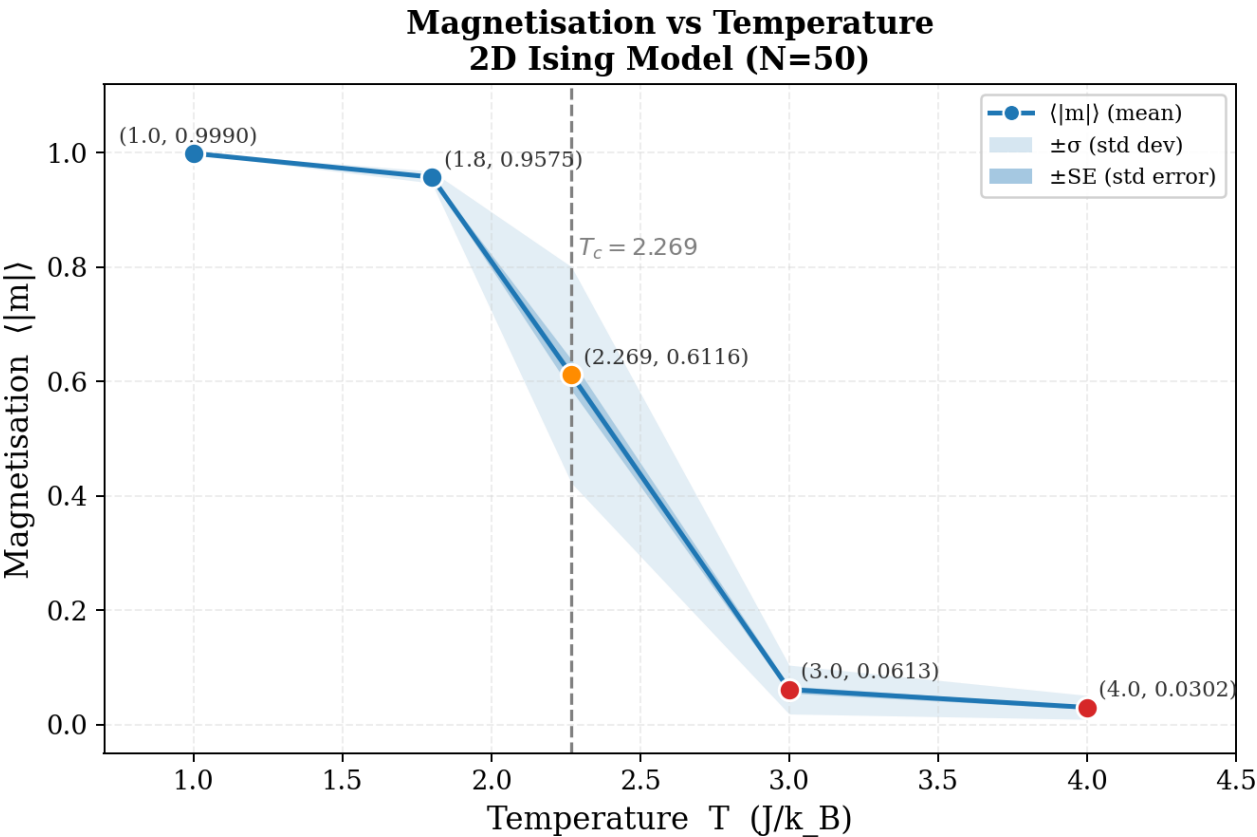
- **Magnetisation** — how aligned the spins are on average
- **Spatial Correlation $G(r)$** — how much two spins influence each other over distance
- **Time Correlation $C(t)$** — how much the system remembers its past state over time
- **Structure Factor $S(k)$** — the spin arrangement pattern in frequency space
- **Cluster Statistics** — how spins group together into connected regions

The shaded bands in each plot show the uncertainty in the results — the wider band is the standard deviation ($\pm\sigma$) and the narrower band is the standard error ($\pm SE$).

Simulation Parameters

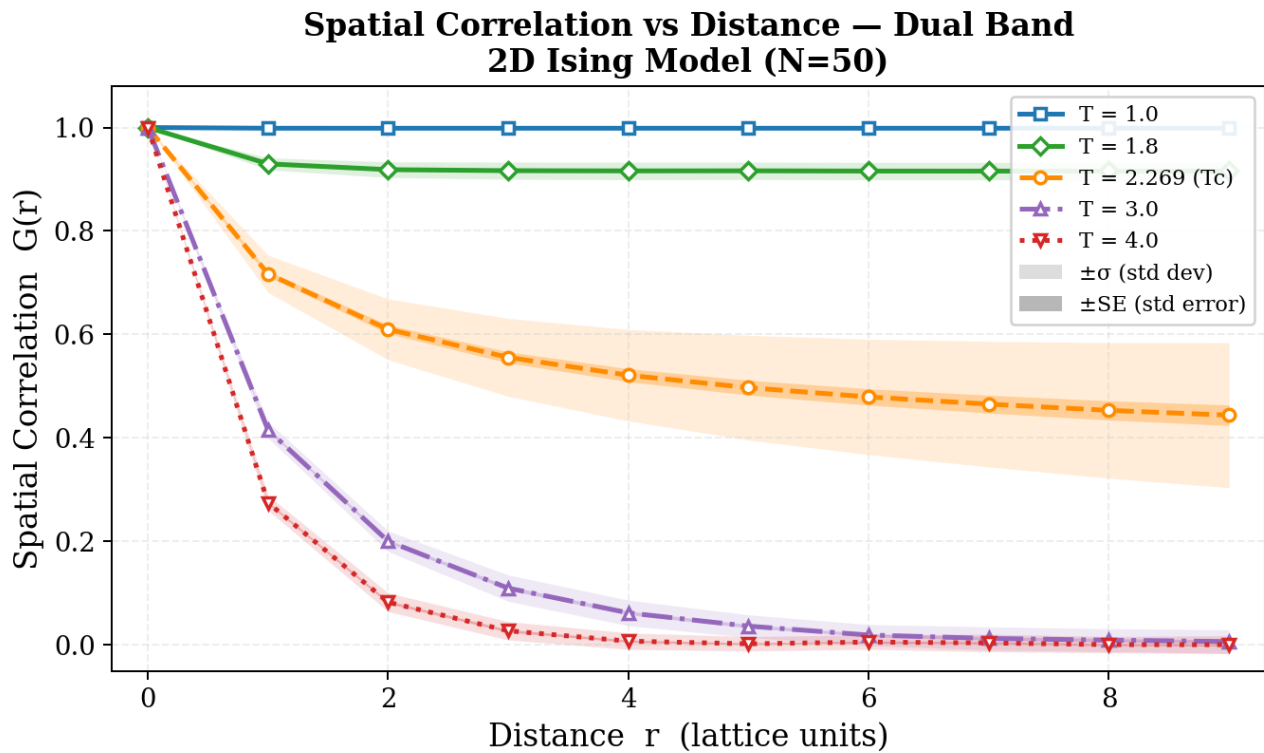
Parameter	Value
Lattice Size	50×50
Algorithm	Metropolis Monte Carlo
Boundary Condition	Periodic Boundary Condition
Equilibration Steps	100,000 MC steps
Number of Samples	50
Measurement Gap	1,000 MC steps
Temperatures Studied	1.0, 1.8, 2.269, 3.0, 4.0
Critical Temperature	$T_c = 2.269$

Figure 1 — Magnetisation vs Temperature



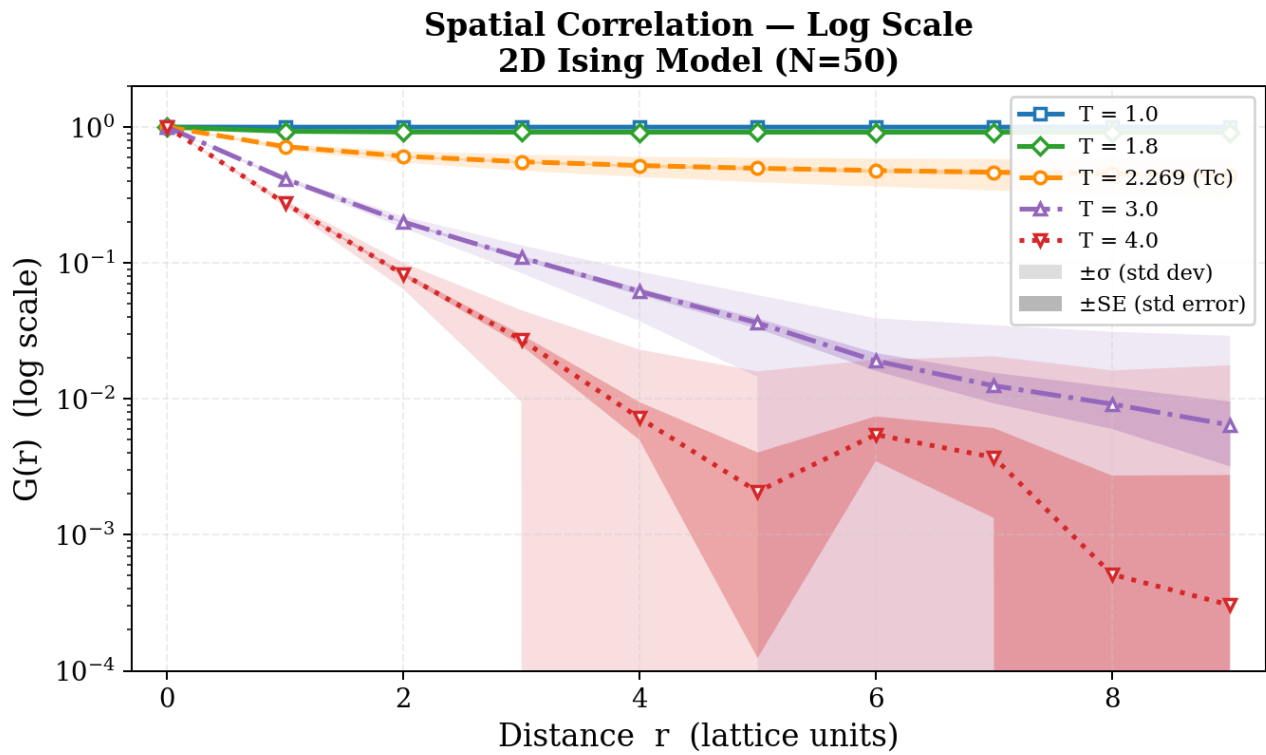
This graph shows how the average magnetisation changes with temperature. At low temperatures, most of the spins point in the same direction, so the magnetisation is high. As the temperature increases, thermal fluctuations increase and the magnetisation decreases. Near the critical temperature $T_c = 2.269$, the magnetisation falls rapidly, and above the critical temperature the system becomes disordered.

Figure 2 — Spatial Correlation Function $G(r)$



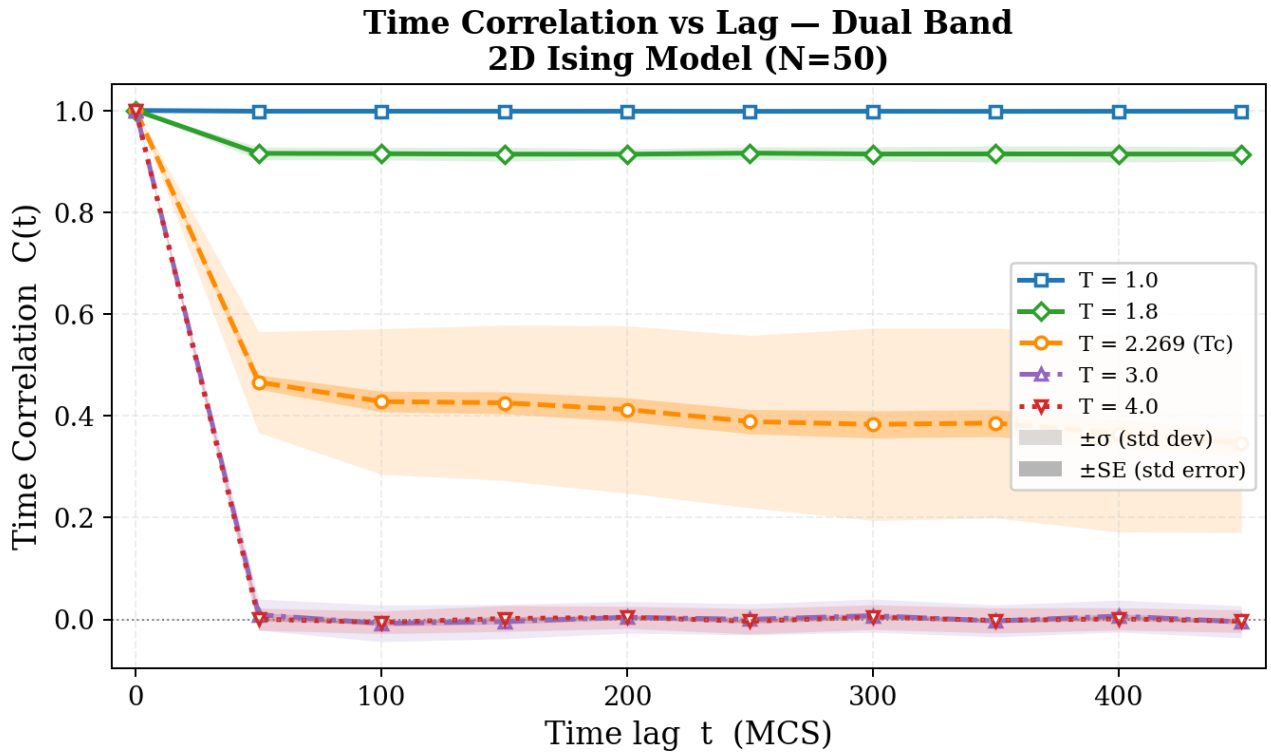
The spatial correlation function tells us how strongly two spins separated by a distance r are related to each other. At low temperatures, spins remain correlated even at large distances because the system is ordered. Near the critical temperature, the correlation decreases slowly. At higher temperatures, the correlation becomes small because the spins become randomly oriented.

Figure 3 — Spatial Correlation (Log Scale)



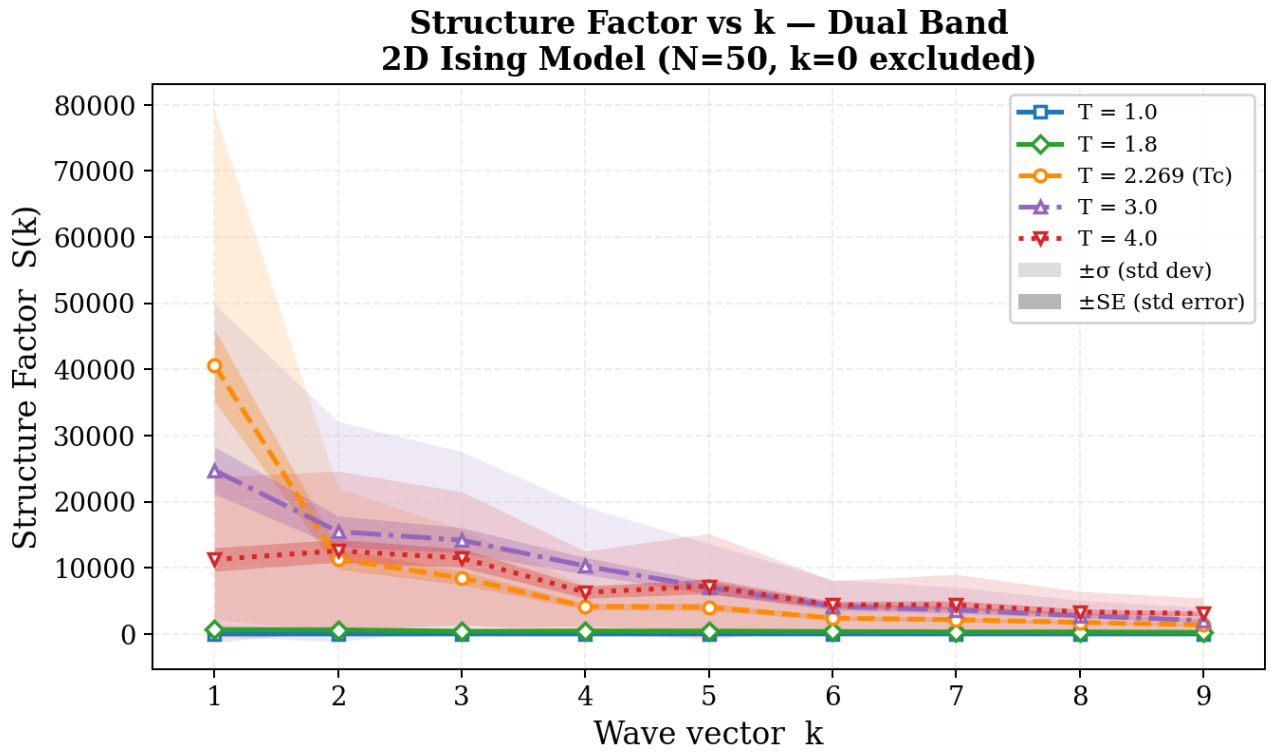
The logarithmic plot helps us see the decrease of correlation more clearly. The correlation becomes very small at large distances, especially at high temperatures. The larger error bands at large distances occur because the measured values become very small and fluctuate more between different samples.

Figure 4 — Time Correlation Function $C(t)$



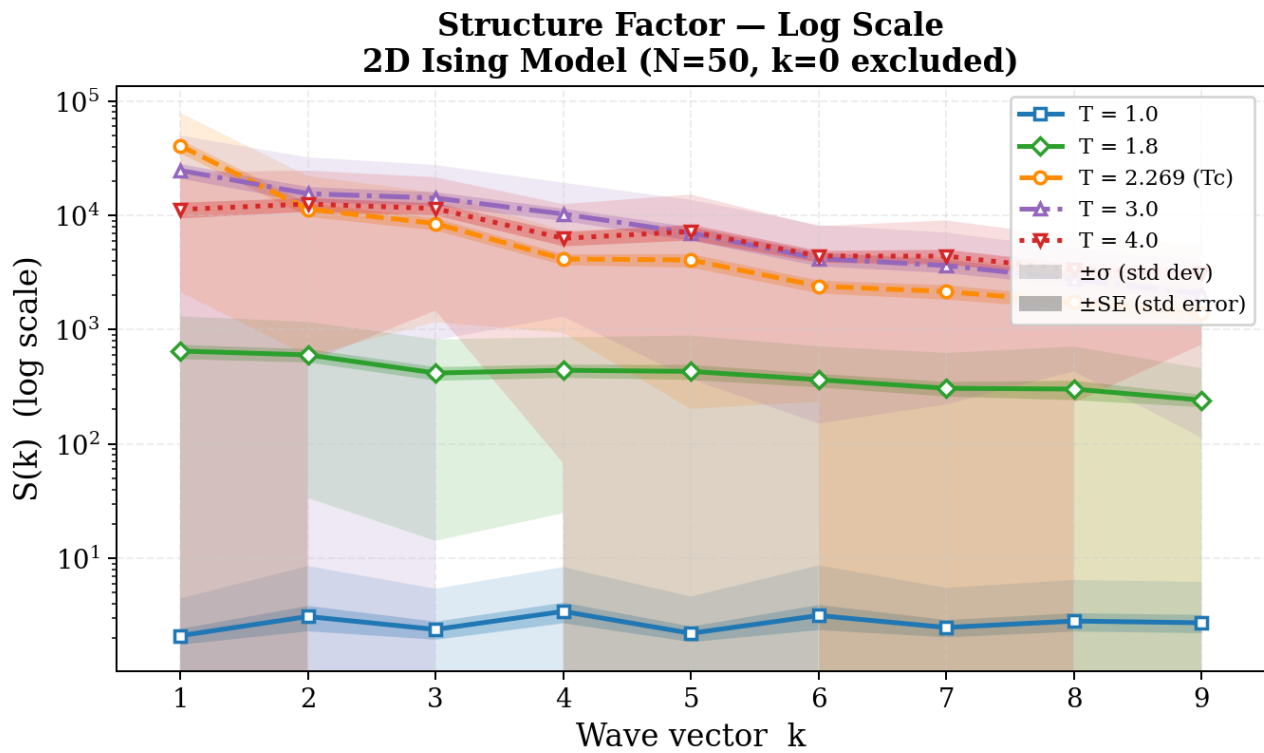
The time correlation function shows how long the system remembers its initial configuration. At low temperatures, the system changes very slowly and the correlation remains high. Near the critical temperature, the decrease becomes slower due to strong fluctuations. At high temperatures, the system loses memory quickly because the spins flip more frequently.

Figure 5 — Structure Factor $S(k)$



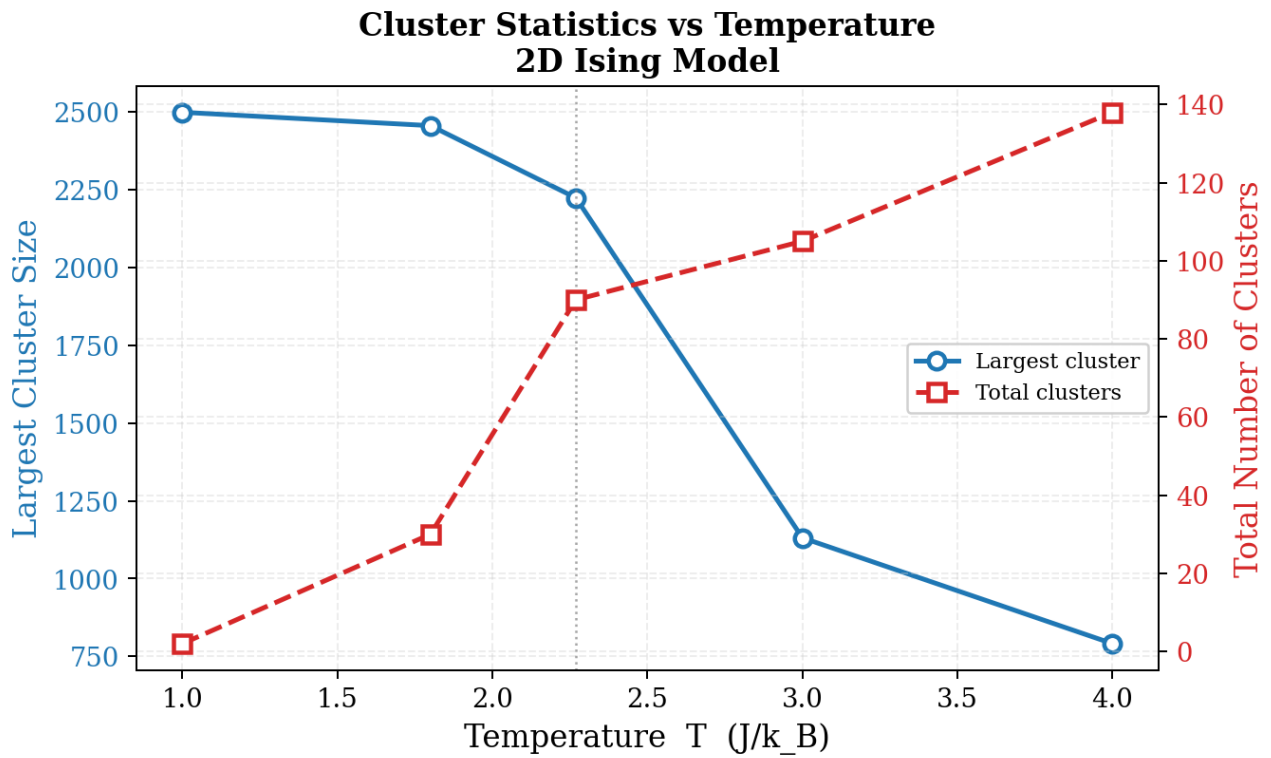
The structure factor gives information about how the spins are arranged inside the lattice. Large values at small values of k indicate the presence of large spin domains. Near the critical temperature, fluctuations become stronger and affect several values of k . At high temperatures, the structure factor becomes smaller because the system becomes disordered.

Figure 6 — Structure Factor $S(k)$ (Log Scale)



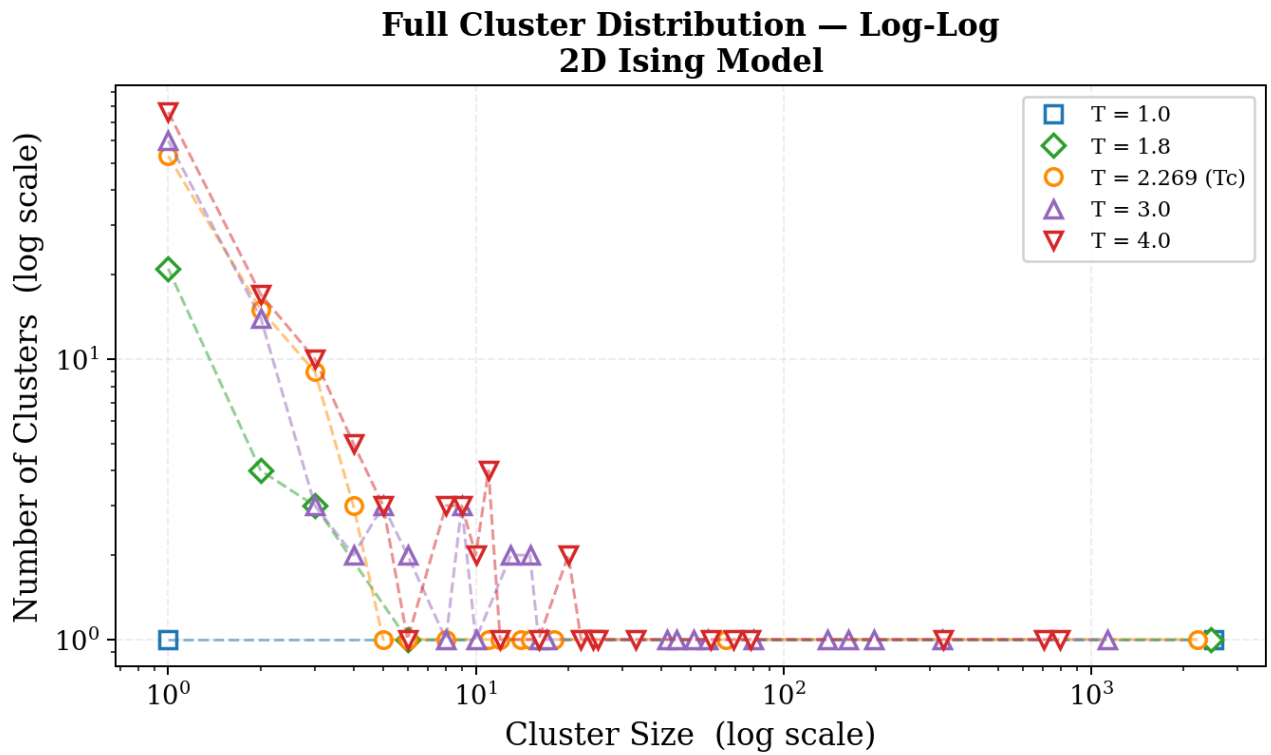
The logarithmic plot of the structure factor makes the smaller values easier to observe. It helps compare the behaviour of the system at different temperatures and shows how the ordering changes as the temperature increases.

Figure 7 — Cluster Statistics vs Temperature



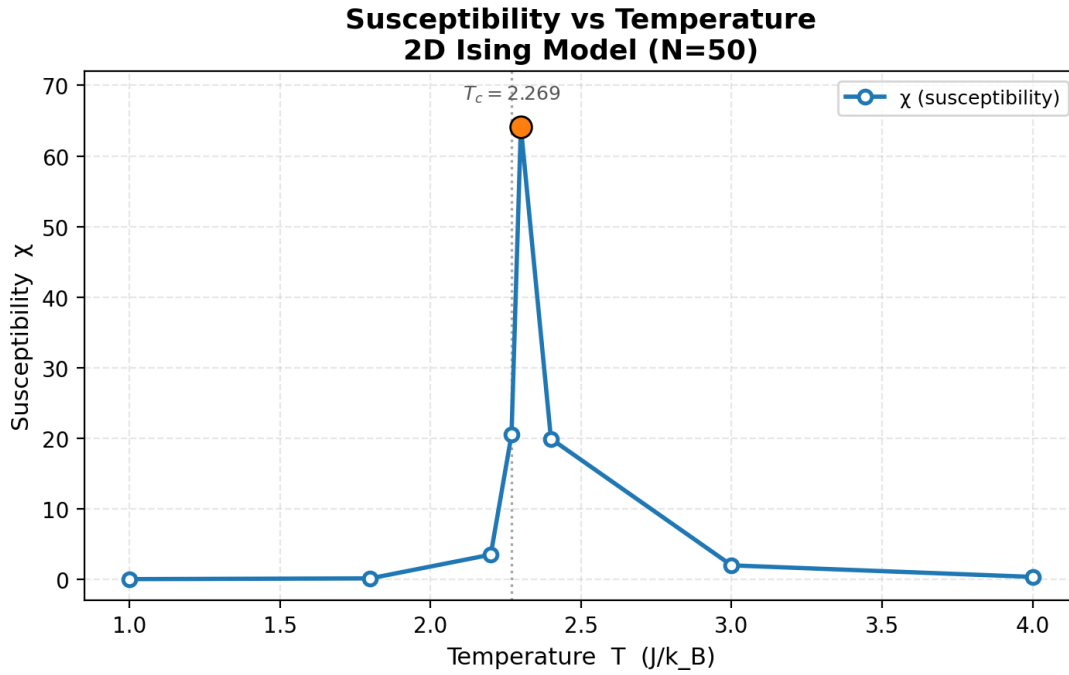
This graph shows how the size of the largest cluster and the total number of clusters change with temperature. At low temperatures, one large cluster occupies most of the lattice. Near the critical temperature, the large cluster starts to break apart and the total number of clusters increases. At high temperatures, many small clusters are present.

Figure 8 — Full Cluster Distribution (Log-Log)



This log-log plot shows the full range of cluster sizes across all temperatures. At low temperatures, almost all spins belong to one giant cluster. Near T_c , a wide range of cluster sizes appears — a typical sign of a phase transition. At high temperatures, many small clusters are present and the large connected regions disappear.

Figure 9 — Susceptibility vs Temperature



Magnetic susceptibility was calculated from the fluctuations in magnetization using 50 ensemble measurements at each temperature. The susceptibility remains very low at low temperatures, increases sharply near the critical region, and reaches a maximum at $T \approx 2.3$. Beyond the critical temperature, it decreases as the system enters the disordered phase. The observed peak is close to the exact critical temperature of the two-dimensional Ising model ($T_c = 2.269$), with the slight shift arising from finite-size effects ($L = 50$).